

FakeAgent / SectopRAT

Signed-application DLL sideloading through renamed CEF and IBM SPSS binaries

CLASSIFICATION	HUNTING
CONFIDENCE	HIGH
PRIMARY PLATFORM	MICROSOFT DEFENDER XDR
RESEARCH DATE	23 JULY 2026

A source-grounded detection study derived from Huntress incident observations across at least 29 organizations. This dossier documents what the available telemetry can observe, what it cannot prove, and how to validate the hunting logic before promotion.

Executive assessment

ASSESSMENT The process-module pairs are campaign-specific and directly supported by primary research. They are appropriate for a high-confidence hunt, but a match does not independently establish that the loaded DLL is malicious.

Dimension	Assessment	Operational consequence
Source quality	Primary Huntress SOC research; direct binary analysis; activity observed on 21-22 July 2026.	Strong basis for a campaign-derived hunting hypothesis.
Observable behavior	Renamed signed executables load co-located DLLs: ClaudeDesktop.exe / DockerDesktop.exe -> libcef.dll; sslconf.exe -> tempdir.dll.	DeviceImageLoadEvents can observe the actor process and loaded image relationship.
Durability	Higher than isolated hashes or domains, but dependent on the attacker retaining these process-module pairings.	Retain as campaign-specific logic; do not present it as a universal DLL sideloading detector.
Promotion state	Not production-validated.	Baseline, file provenance, signer/original-name analysis and prevalence are required.

Decision

Publish as: High-confidence hunting query / untested implementation sketch. The query intentionally avoids prevalence thresholds, hard-coded paths, campaign IP addresses and blockchain indicators.

EVIDENCE BOUNDARY The source supports the pairing and campaign context. The query does not inspect DLL contents, validate hashes, prove tampering, or confirm SectopRAT execution.

Attack study



Observed execution branches

Branch	Signed host process	Loaded module	Research-supported role
CEF branch	ClaudeDesktop.exe / DockerDesktop.exe; reported as renamed copies of JetBrains jcef_helper.exe.	libcef.dll	Initial staging and recurring reinfection. DockerDesktop.exe was associated with scheduled-task persistence.
IBM SPSS branch	sslconf.exe located beneath a user AppData path.	tempdir.dll	Second sideloading path with GPU anti-VM checks and shader-based payload decryption.

Analytical separation

Category	Statement
Source fact	Huntress observed the named process-module pairs and analyzed the corresponding binaries.
Source interpretation	The final payload attribution was updated to SectopRAT after analysis of HVNC functionality and C2 services.
Detection inference	The pair is more discriminating than either the process filename or DLL filename alone.
Assumption	Microsoft Defender for Endpoint collects the relevant DLL-load events without material gaps or sampling.
Unconfirmed by the query	Whether the loaded DLL is modified, malicious, packed, or followed by successful C2.

Telemetry and evidence model

PRIMARY TELEMETRY

Microsoft Defender XDR - DeviceImageLoadEvents. This table is selected because it records DLL loads and exposes both the initiating process and loaded image in the same event.

Field	Use in the hunt	Validation note
Timestamp / DeviceId / DeviceName	Event timing and endpoint correlation.	Confirm retention and stable DeviceId coverage.
InitiatingProcessFileName	Identifies the signed host process name.	Filename alone is not malicious.
InitiatingProcessFolderPath	Locates the host executable.	Use for triage; do not hard-code until baselined.
FileName / FolderPath	Identifies and locates the loaded module.	Recover the file and validate signer, hash and original name.
SHA1 / InitiatingProcessSHA1	Supports prevalence and artifact comparison.	Hashes enrich the investigation; they are not the durable detection condition.
ReportId	Supports event uniqueness with DeviceName and Timestamp.	Preserve for repeatable investigation.

Visibility boundary

Visible in the selected event	Requires enrichment or another table	Not established by telemetry alone
Process name; process path; loaded DLL name and path; hashes; device and time.	Original filename; certificate trust; file creation/download origin; scheduled task; process/network follow-on; organization-wide prevalence.	Malicious file contents; successful sideload execution; SectopRAT attribution; credential theft; C2 success.

Detection engineering

Property	Value
Status	Hunting query - Untested implementation sketch
Primary platform	Microsoft Defender XDR Advanced Hunting
Trigger	Either of the two research-supported process-module pairings.
No threshold	A single event is returned for investigation; volume must be established through baseline.
Deliberate exclusions	No campaign domain, IP, hash, signer, path, prevalence threshold or C2 indicator is required.

```
DeviceImageLoadEvents
| where
  (InitiatingProcessFileName in~ ("ClaudeDesktop.exe", "DockerDesktop.exe")
   and FileName =~ "libcef.dll")
  or
  (InitiatingProcessFileName =~ "sslconf.exe"
   and FileName =~ "tempdir.dll")
| project
  Timestamp,
  DeviceId,
  DeviceName,
  InitiatingProcessFileName,
  InitiatingProcessFolderPath,
  InitiatingProcessCommandLine,
  FileName,
  FolderPath,
  SHA1,
  InitiatingProcessSHA1,
  ReportId
```

WHY THESE CONDITIONS EXIST

Each process-module pair is documented by the primary source. The hunt combines both objects because generic libcef.dll or signed-process activity would create substantial benign overlap.

Triage and false-positive control

Priority	Analyst action	Decision value
1	Recover process and module hashes; inspect original filenames, signature state and certificate chain.	Tests masquerading and tampering hypotheses.
2	Compare process and DLL directories; establish whether they were recently written beneath user-writable or extracted locations.	Supports or weakens the sideloading scenario.
3	Measure tenant prevalence by hash, device, path and software owner.	Separates isolated artifacts from approved packaging.
4	Review file creation, scheduled-task creation, child processes and network activity on the same DeviceId.	Looks for the broader FakeAgent execution chain.
5	Escalate only when provenance, path, hash, persistence or follow-on behavior remains unexplained.	Prevents a filename-only verdict.

False-positive matrix

Scenario	Expected signal	Recommended handling
Internal CEF packaging	Authorized software may rename or wrap CEF components and load libcef.dll.	Validate software owner, known package hash, install source and standard deployment path.
Legitimate IBM SPSS	Signed sslconf.exe may load an authentic tempdir.dll.	Confirm product installation, DLL signature/hash and vendor-owned path.
Malware-analysis lab	Known samples may deliberately reproduce the pairing.	Scope by laboratory assets and preserve visibility; avoid broad suppression.

DO NOT ALLOWLIST Do not suppress all libcef.dll loads, all signed binaries, all AppData execution, or every host used for software testing. Tuning should be evidence-based and as narrow as possible.

Validation and promotion gate

Phase	Method	Exit criterion
Telemetry check	Confirm DeviceImageLoadEvents population and required fields across representative endpoints.	Reliable image-load coverage and non-empty actor/module fields.
30-90 day baseline	Run the hunt and enumerate distinct process path, module path, hashes, devices and owners.	Null or low volume; benign cases separable using stable attributes.
Artifact verification	Inspect original filenames, signatures, hashes, creation source and package ownership.	Unexplained or tampered artifacts can be escalated consistently.
Correlation test	Review process, file, scheduled-task and network events around each match.	Related behaviors are observable and materially strengthen fidelity.
Promotion review	Define narrow suppressions and simulate the candidate analytic rule.	Acceptable alert volume and documented response playbook.

Outcome	Condition
Promote	Matches are absent or rare, benign cases are cleanly distinguishable, and telemetry coverage is sufficient.
Keep as hunting	The pairing remains useful for periodic research but context is needed before escalation.
Reject	Authorized applications generate the same pairs at scale without stable discriminators.
Telemetry failure	Image-load telemetry is missing, sampled, or does not preserve the actor/module relationship.

MITRE ATT&CK; mapping

Technique	Mapping rationale
T1574.002 - DLL Side-Loading	A signed executable loads a maliciously modified DLL from its search path.
T1036 - Masquerading	JetBrains components are renamed to impersonate Claude Desktop and Docker Desktop.
T1053.005 - Scheduled Task	The source reports DockerDesktop.exe persistence through a scheduled task; the current query does not detect task creation.

References and review notes

Primary research

Huntress. Inside FakeAgent: How a Claude Desktop Malvertising Campaign Hit 29 Organizations with SectopRAT. Published 22 July 2026. Activity observed 21-22 July 2026.
<https://www.huntress.com/blog/fakeagent-claude-desktop-malvertising-ends-in-dotnet-rat>

Official telemetry documentation

Microsoft Learn. DeviceImageLoadEvents table in the Microsoft Defender XDR advanced hunting schema.
<https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceimageloadevents-table>

Technique references

Reference	Purpose
MITRE ATT&CK; T1574.002	DLL Side-Loading behavior.
MITRE ATT&CK; T1036	Masquerading behavior.
MITRE ATT&CK; T1053.005	Scheduled-task persistence reported by the source.

REVIEW STATUS

This package is research-derived and has not been executed against a target tenant. Query syntax and documented fields have been reviewed; operational fidelity remains dependent on local telemetry and baseline.

Quality-control checklist

Control	Status
Primary source materially reviewed	Completed
Official schema reviewed	Completed
Customer identifiers and internal IOCs removed	Completed
Query based on directly observable behavior	Completed
Production validation	Not completed - hunting only